

Daily Progress Report Day 6

Project Name: MediFlow CRM — Doctor / Clinic Practice Management System

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Reporting Period: Daily

1. Summary of Today's Activities

Today, the team focused on resolving critical operational bottlenecks, accelerating platform performance, refining clinical scheduling workflows, and executing a complete, end-to-end visual overhaul of **MediFlow CRM** into a **Premium Apple-Inspired UI System**. Specifically, we resolved the route exception on the practice Reports module and delivered a full-featured multi-domain analytics engine with CSV and printable exports. We eliminated application latency via SQLite Write-Ahead Logging (WAL) and hover link prefetching, streamlined physician assignment logic in the appointment workflow, and comprehensively revamped every Blade template across the entire platform with soft glassmorphism, refined typography, and mobile-first ergonomics while maintaining a 100% test pass rate (98/98 unit & feature tests).

2. Key Discussions & Decisions

A. Reports & Analytics Engine Implementation & Bug Fix

- **Error Resolution & Core Controller:** Resolved the missing route / 500 error triggered by the practice navigation by creating and registering ReportController.php mapped to /reports.
- **Four-Domain Practice Analytics:** Implemented comprehensive analytical aggregation across:
 1. **Financial & Revenue:** Gross billings, collected payments, outstanding balances, tax liabilities, and discount metrics.
 2. **Appointments & Scheduling:** Total booked encounters, completion rates, no-show rates, cancellation volume, and per-doctor workload.
 3. **Patient Demographics:** Total registered base, new patient acquisition rate, gender distribution, and age tier categorization.
 4. **Prescriptions & Clinical Encounters:** Prescriptions issued, consultation volume, and top-prescribed pharmaceuticals.
- **Exporting & Print Engine:** Built dynamic CSV streaming export (/reports/export) for accounting audits and an A4-optimized printable clinical report template (/reports/print) styled with clean medical print typography.

B. Performance Optimization & Page Transition Acceleration

- **Database Concurrency & WAL Mode:** Addressed SQLite table locking and transaction latency by configuring Write-Ahead Logging (`PRAGMA journal_mode = WAL;`) and relaxed synchronization (`PRAGMA synchronous = NORMAL;`), enabling non-blocking concurrent reads and instantaneous database writes.
- **Instant Hover-Intent Prefetching:** Integrated lightweight client-side prefetching in `app.blade.php` that warms the browser cache on link hover/focus, producing near-zero perceived page transition times.
- **Asset Loading & Session Tuning:** Reconfigured Google Fonts with asynchronous loading (`display=swap`) to eliminate render-blocking CSS, optimized Vite bundling, and switched session storage to high-speed file caching.

C. Appointment Workflow & Doctor Assignment Simplification

- **Nomenclature & Clinical Scoping:** Removed confusing naming conventions ("doctor prescription app") and replaced them with standard "Doctor" designations. Scoped doctor queries strictly to active accounts with the `admin_doctor` role.
- **Smart Conditional Doctor Selection:** Implemented intelligent UX logic within `AppointmentController.php` and appointment views:
 - **Single-Doctor Practices:** Automatically pre-assigns the sole practitioner with a sleek read-only physician badge, eliminating redundant dropdown clicks.
 - **Multi-Doctor Practices:** Dynamically presents a clear doctor selection dropdown allowing staff to assign specific consulting physicians.

D. Visual Redesign — Premium Apple-Inspired Glassmorphic UI System

- **Design System Foundation:** Overhauled the visual presentation layer into a calm, minimal, Apple/iOS-inspired aesthetic using a soft cool neutral palette (`#F4F7F8`, `#EEF5F6`), soft cyan/teal accents (`#318FAA`, `#2C91AE`), and high-contrast slate typography (`#11181C`).
- **Glassmorphic Component Architecture:** Built reusable design utility classes (`.glass-card`, `.glass-card-elevated`, `.glass-dock`, `.apple-input`, `.btn-pill-primary`, `.glass-pill`) featuring translucent frosted backgrounds (`backdrop-blur-md bg-white/75 border-white/60`), subtle depth shadows, and generous rounded contours (`rounded-[24px]` and `rounded-[28px]`).
- **Omnipresent Mobile Glass Dock:** Replaced traditional mobile menus with an iOS-style floating bottom glass dock providing thumb-friendly navigation across core clinic modules.
- **Universal View Overhaul:** Completely revamped 15+ Blade templates across the platform:
 - Master Layouts & Navigation (`app.blade.php`, `navigation-items.blade.php`, `guest.blade.php`)
 - Practice Cockpit Dashboard (`dashboard.blade.php`)
 - Appointments System (`appointments/index.blade.php`, `create.blade.php`, `show.blade.php`, `edit.blade.php`)

- Patient Hub & 9-Section Profiles
(patients/index.blade.php, create.blade.php, show.blade.php, edit.blade.php)
- Digital Prescriptions Hub (prescriptions/index.blade.php, create.blade.php, show.blade.php)
- Invoices & Billing Operations (invoices/index.blade.php, create.blade.php, show.blade.php)
- Reports Dashboard & Printable Output (reports/index.blade.php, reports/print.blade.php)
- Practice Settings & Subaccounts Directory
(settings/index.blade.php, subaccounts/index.blade.php)
- Global Spotlight Search & Notifications (partials/spotlight-search.blade.php, notifications/index.blade.php)

3. Members Role

Team Member	Assigned Role	Module & Responsibility Breakdown
Anwasha Manna (Backend & Database)	Backend Architecture & Database Lead	• Resolved the /reports route failure and engineered ReportController.php analytical aggregation queries (Revenue, Appointments, Patients, Medications) • Implemented backend CSV streaming export with dynamic header generation and automated filtering • Tuned database performance by configuring SQLite Write-Ahead Logging (WAL) and synchronous normal mode • Refactored user role filtering in AppointmentController.php to strictly isolate active admin_doctor accounts
Indranil Dutta (Full-Stack / Clinical Workflow Lead)	Clinical Engine & Workflow Specialist	• Engineered the conditional doctor assignment workflow (single-doctor auto-assignment badge vs. multi-doctor selector) • Redesigned the Appointment creation and scheduling interfaces with Apple-inspired floating form containers • Updated the Digital Prescription builder and Clinical Consultation encounter views with glass cards and pill action buttons • Designed the clean, print-friendly A4 layout for the printable medical report view (reports/print.blade.php)
Debarati Gantait (Frontend, Billing & Reporting Lead)	Billing, Analytics & UI/UX Lead	• Conceptualized and implemented the system-wide Apple glassmorphic design architecture across 15+ Blade templates • Designed the floating mobile glass dock (.glass-dock) and responsive desktop navigation with cyan glow accents • Implemented client-side hover prefetching and

		asynchronous font delivery to eliminate page transition delays • Revamped the Invoicing, Billing, and Analytics dashboard UI with frosted KPI stat cards, badges, and financial breakdowns
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4. Roadblocks & Knowledge Gathering Strategy

- **Area of Focus:** Eliminating page navigation sluggishness caused by database locking, resolving the missing reports crash, and executing a complete visual overhaul across 15+ complex Blade templates without breaking any existing business logic, validation rules, or authorization gates.
 - **Action Taken:** Activated SQLite Write-Ahead Logging (WAL) to decouple read/write locks and implemented instant hover prefetching in the master layout; built a robust multi-domain ReportController to power analytics and CSV exports; and implemented the Apple design system via modular Tailwind CSS classes while preserving all existing routes, form actions, and permission middleware. Verified full stability by running the complete automated test suite with a 100% pass rate (**98/98 tests passed, 376 assertions**).
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Submitted By GROUP A

(Anwesha Manna, Debarati Gantait, Indranil Dutta)